An Entity Matching-Based Modular Framework for Numerical Consistency Verification Across Heterogeneous Construction Progress Payment Documents

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Abstract

Construction progress payment verification requires numerical consistency checks across multiple cost-related documents with heterogeneous structures and different aggregation levels. These differences complicate identification of corresponding work items and their numerical relationships. This study evaluates a hierarchy-aware numerical consistency verification approach that represents document context, work-item attributes, and parent–child hierarchies to construct 1:1 and 1:N verification relationships. Relation-specific constraints are then applied to quantity equivalence, line-item arithmetic, and hierarchical aggregation, with source references retained for traceable verification. A pilot experiment used earthwork quantity takeoff and bill-of-quantities records from a single construction project. The evaluation comprised 96 relation candidates and 32 verification cases, including 24 original and eight controlled-perturbation cases. An ablation study compared approaches with and without work-item hierarchy. Incorporating hierarchy improved overall relation identification F1 from 0.857 to 0.909, and exact reconstruction of four independent multi-child aggregation groups from 3/4 to 4/4. Verification coverage increased from 80.6% to 87.1%, and three-class accuracy from 81.3% to 87.5%. Cross-document quantity matching remained unchanged (F1 = 0.714) in both configurations. These preliminary findings indicate that work-item hierarchies can improve aggregation-relation construction and extend numerical verification coverage, while highlighting the need to evaluate broader document variation in future work.

1. Introduction

- Construction progress payments require verification of quantities, unit prices, amounts, and cumulative payment records across multiple cost-related documents.
- However, heterogeneous document structures, inconsistent item representations, and different aggregation levels make it challenging to identify corresponding items and verify numerical relationships across documents.
- Although previous studies have explored construction cost data extraction, payment automation, and rule-based cost verification, the applicability of existing approaches to numerical consistency verification across heterogeneous progress payment documents remains insufficiently examined.
- Therefore, this study aims to adapt and evaluate a numerical consistency verification method for earthwork progress payment documents, focusing on detecting arithmetic and aggregation discrepancies and providing traceable verification evidence through a pilot experiment.

2. Proposed Method

- This study proposed a hierarchy-aware numerical consistency verification approach for cost-related information distributed across heterogeneous construction progress payment documents.
- Numerical records are structured with document context, work-item hierarchies, and corresponding item relationships to enable cross-document verification.
- Relation-specific numerical constraints are selected and executed to verify arithmetic calculations, equivalent values, and hierarchical aggregation relationships.
- Numerical inconsistencies are identified by comparing recorded and calculated values, while preserving traceable evidence of discrepancies.

A | Quantity Takeoff

Work item	Q(m³)
Fill / subgrade	4,040.8
Fill / subbody	33,560.356
Cut / soil	182.2

Measured Q (m³)

B₁ | Detailed BOQ

Work item	Q	U	Cost
Subgrade fill	4,041	1,239	5,006,799
Subbody fill	33,560	900	30,204,000
Soil cut	182	645	117,390

Q: m³ U: KRW/m³ Cost: KRW

B₂ | Work Summary

Earthwork total 1,034,903,088 KRW

Fill	**35,211,299**
Cut	117,390
Excavation	92,405,642
Backfilling	62,009,604
Soil transport	845,159,153

CROSS-DOCUMENT / CROSS-TABLE NUMERICAL RELATIONS
A ↔ B₁: corresponding quantities | B₁ (2 fill items) → B₂ (Fill): 1:N

NUMERICAL CONSISTENCY VERIFICATION

(i) Quantity equivalence

$\Delta Q_i = |Q_i^{B_1} - \text{round}(Q_i^{A})|$

4,041 = round(4,040.8)

PASS (ΔQ = 0 m³)

(ii) Arithmetic consistency

$\Delta A_i = |A_i - Q_i \times U_i|$

5,006,799 = 4,041 × 1,239

PASS (ΔA = 0 KRW)

(iii) Hierarchical aggregation

$\Delta A_p = |A_p^{B_2} - \sum_{c \in C(p)} A_c^{B_1}|$

35,211,299 − 35,210,799 = 500

FAIL (ΔA = 500 KRW)

DECISION RULE ΔQ = 0 → PASS
ΔA = 0 → PASS | ΔA > 0 → FAIL

NOTATION i: item; p: parent; c: child
C(p): child set of p; Q: quantity; U: unit rate; A: amount

Fig. 1. Proposed framework for hierarchy-aware cross-document numerical consistency verification.

3. Pilot Experiment & Results

A. Pilot Experimental Setup

1	2	96	32
Project	Workbooks	Relation candidates	Verification cases

Earthwork only | 9 BOQ items | 24 original + 8 controlled perturbation cases

B. Hierarchy Ablation Results

Overall Numerical Relation Identification (F1-score)



Disjoint 1:N Child-set Reconstruction



- Hierarchy recovered one additional independent multi-child aggregation group; cross-document quantity F1 remained 0.714 for both configurations.

Fig. 2. Impact of hierarchy on relation identification and 1:N child-set reconstruction.

C. Numerical Verification Results

Metric	Without Hierarchy	With Hierarchy
Verification Coverage	25/31 (80.6%)	**27/31 (87.1%)**
Three-class Accuracy	26/32 (81.3%)	**28/32 (87.5%)**

Table 1. Numerical verification outcomes with and without hierarchy information.

Key Finding

- Work-item hierarchy extended 1:N aggregation-relation construction and increased the proportion of num erically verifiable cases.

4. Conclusion

- This study evaluated hierarchy-aware numerical consistency verification using earthwork quantity takeoff and bill-of-quantities documents.
- Incorporating work-item hierarchy improved numerical relation identification (F1: 0.857→0.909) and increased verification coverage from 80.6% to 87.1%.
- The results suggest that hierarchical information can facilitate the reconstruction of 1:N aggregation relationships, extending the applicability of numerical consistency verification across construction cost tables.
- Future research will investigate the applicability of the proposed approach to more diverse progress payment documents and construction projects.

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